

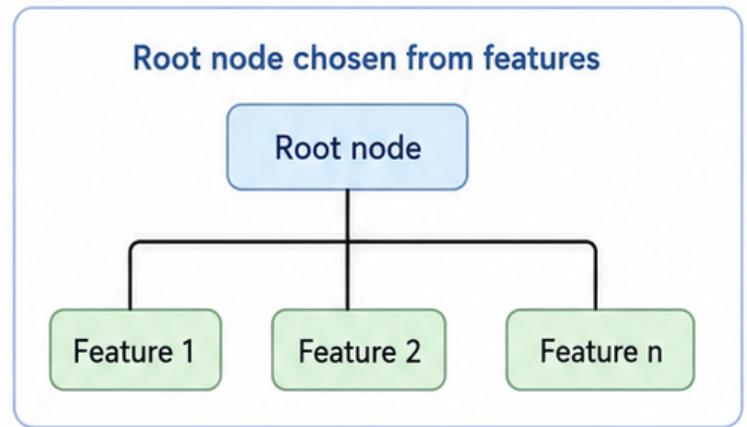
Decision Tree Algorithm



Introduction, entropy, and sample dataset

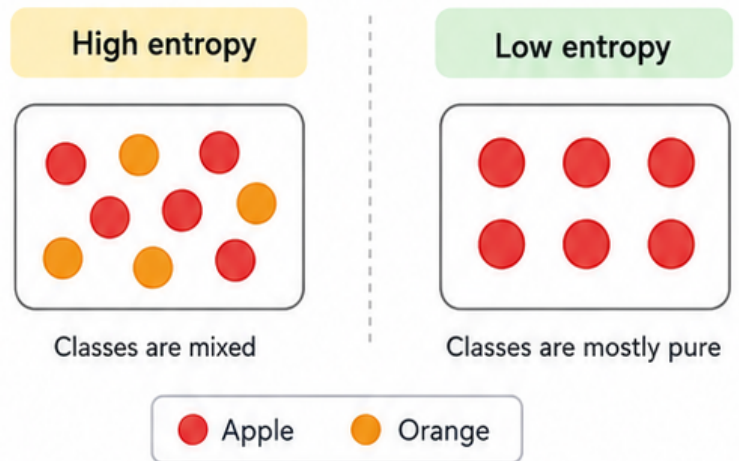
1 What is a decision tree?

- A decision tree splits data based on features to make decisions or predictions.
- The root node is chosen using the best split criterion.



2 Entropy

- Entropy measures impurity, randomness, or disorder in a dataset.
- High entropy means the classes are mixed.
- Low entropy means the classes are mostly pure.



3 Example dataset

Weight	Color	Label
150	8	Apple
170	9	Apple
130	6	Apple
160	3	Orange
120	2	Orange



Features
 $X = [\text{weight, color}]$



Target
 $y = \text{label}$



Total records
 $= 5$



Apples
 $= 3$



Oranges
 $= 2$

Decision Tree Algorithm



Entropy of the dataset and first split candidate

1 Entropy formula

$$H(S) = -p(\text{Apple}) \log_2 p(\text{Apple}) - p(\text{Orange}) \log_2 p(\text{Orange})$$

- $p(\text{Apple}) = 3/5 = 0.6$
- $p(\text{Orange}) = 2/5 = 0.4$

$$H(S) = -(3/5) \log_2(3/5) - (2/5) \log_2(2/5)$$

= 0.971 bits

2 Candidate split: Color ≤ 2.5

Split on Color ≤ 2.5

Left (≤ 2.5)

Right (> 2.5)



Color	Label
2	Orange

Entropy(left) = 0

(Pure subset)

Color	Label
3	Orange
6	Apple
8	Apple
9	Apple

$$H(\text{right}) = -(3/4) \log_2(3/4) - (1/4) \log_2(1/4)$$

= 0.811 bits

3 Weighted entropy

$$\text{Weighted entropy} = (1/5)(0) + (4/5)(0.811) = \mathbf{0.649 \text{ bits}}$$

★ This split reduces impurity, but not perfectly.



Features

$X = [\text{weight}, \text{color}]$



Target

$y = \text{label}$



Total records

= 5



Apples

= 3



Oranges

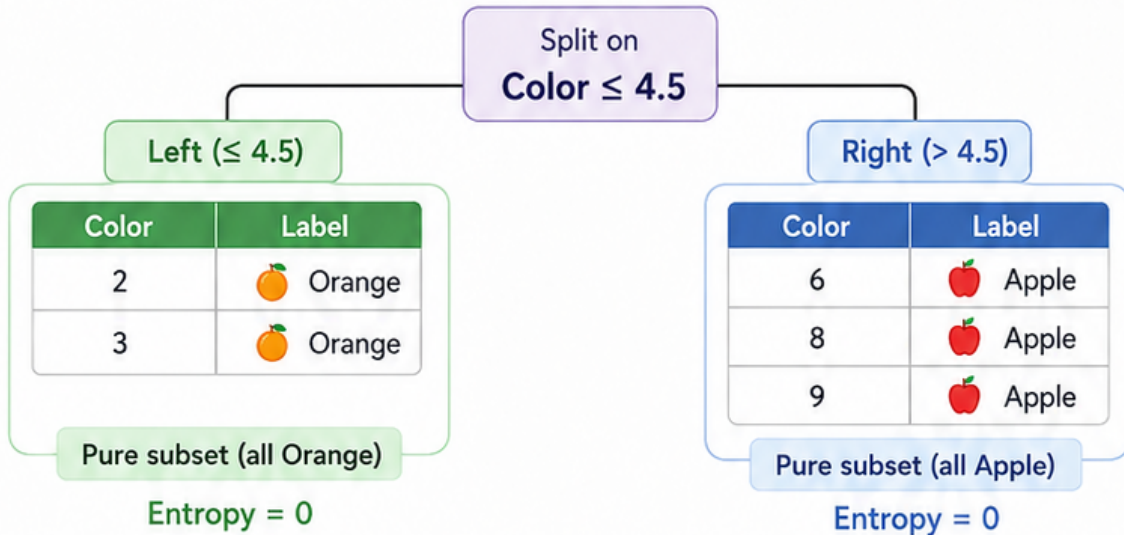
= 2

Decision Tree Algorithm



Evaluating better split candidates

1 Candidate split: Color ≤ 4.5



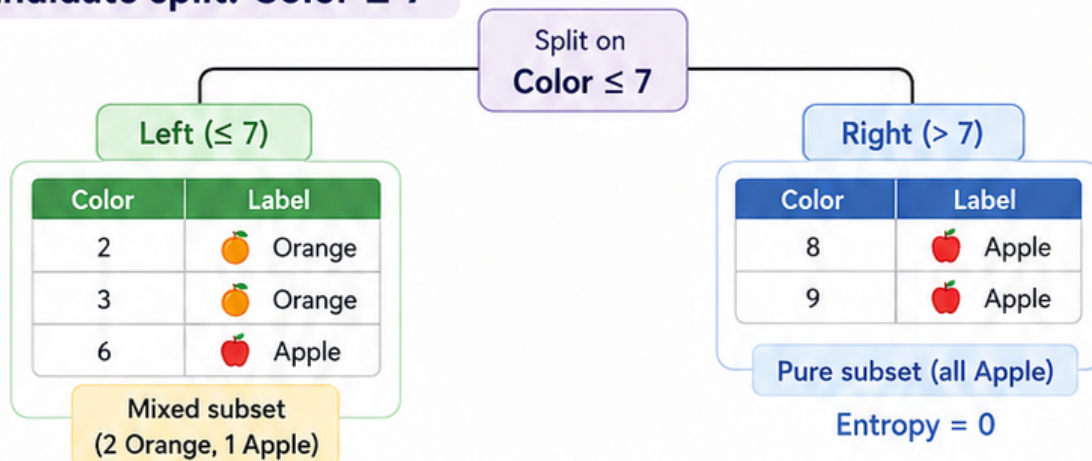
- Weighted entropy = $\frac{2}{5} (0) + \frac{3}{5} (0) = 0$
- Information gain = $0.971 - 0 = 0.971$



Excellent split!

Perfectly separates the classes.

2 Candidate split: Color ≤ 7



- Entropy (left) $H_{\text{left}} = -\frac{1}{3} \log_2 \left(\frac{1}{3} \right) - \frac{2}{3} \log_2 \left(\frac{2}{3} \right) = 0.918$
- Entropy (right) $H_{\text{right}} = 0$
- Weighted entropy = $\frac{3}{5} (0.918) + \frac{2}{5} (0) = 0.551$



Lower than other imperfect choices, but not as good as the split at 4.5.



Features
 $X = [\text{weight}, \text{color}]$



Target
 $y = \text{label}$



Total records
 $= 5$



Apples
 $= 3$



Oranges
 $= 2$

Decision Tree Algorithm



Comparing information gain across split thresholds

1 Candidate split: Color ≤ 7

Weighted entropy calculation

$$\begin{aligned}E_{\text{weighted}} &= \frac{3}{5} (0.918) + \frac{2}{5} (0) \\&= 0.600 + 0.000 \\&= \mathbf{0.551}\end{aligned}$$

Information gain

$$IG = 0.971 - 0.551 = \mathbf{0.420}$$

Left (Color ≤ 7)
3 records



(0.918 entropy)

Right (Color > 7)
2 records



(0.000 entropy)



Better split, but not perfect.

2 Weaker candidate split (higher threshold)

Summary

- Weighted entropy = 0.800
- Information gain = **0.171**



Weak split: leaves are still impure (many classes mixed).

3 Comparison of candidate splits

Threshold	Weighted Entropy	Information Gain	Comment
2.5	0.649	0.322	Good, but not pure
4.5	0.000	0.971	Best split
7.0	0.551	0.420	Better, but not perfect
Higher threshold	0.800	0.171	Weak split



Conclusion

The best root split is **Color ≤ 4.5** because it gives the highest information gain and perfectly separates the classes.



Features
 $X = [\text{weight}, \text{color}]$



Target
 $y = \text{label}$



Total records
 $= 5$



Apples
 $= 3$

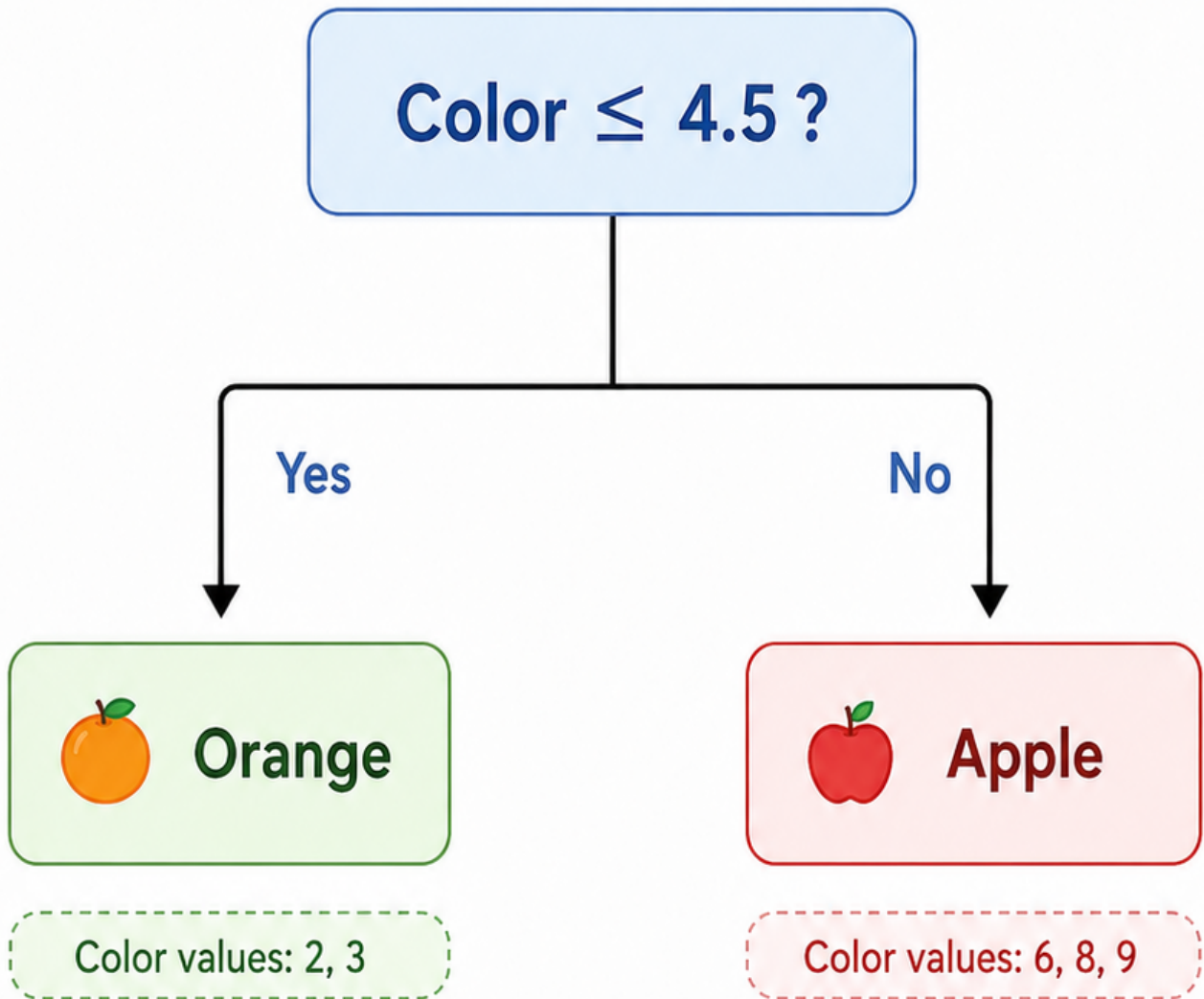


Oranges
 $= 2$

Decision Tree Algorithm



Final decision tree



Because the split at **Color ≤ 4.5** creates two pure groups, it becomes the best root rule for this dataset.

